

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 7 of 10

The First Law of Thermodynamics for Closed Systems

Joule's experiments · the first law for a cycle · energy is a property · the first law for a process · internal energy · PMM1 · limitations of the first law

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Joule's experiments

and the first law for a cycle

Joule's paddle-wheel experiments

- Water in an insulated vessel stirred by a paddle driven by falling weights (1840–49)
- Work done and temperature rise measured
- Water then cooled back to its initial temperature – a complete **cycle**
- Work supplied always proportional to heat rejected, whatever the manner of the work

KEY POINT The mechanical equivalent of heat: $4.186 \text{ kJ} = 1 \text{ kcal}$.

The first law for a cycle

$$\oint \delta Q = \oint \delta W$$

- For a closed system undergoing a cycle, the **net heat transfer equals the net work transfer** (in consistent units)
- Heat and work are mutually convertible – the basis of every heat engine

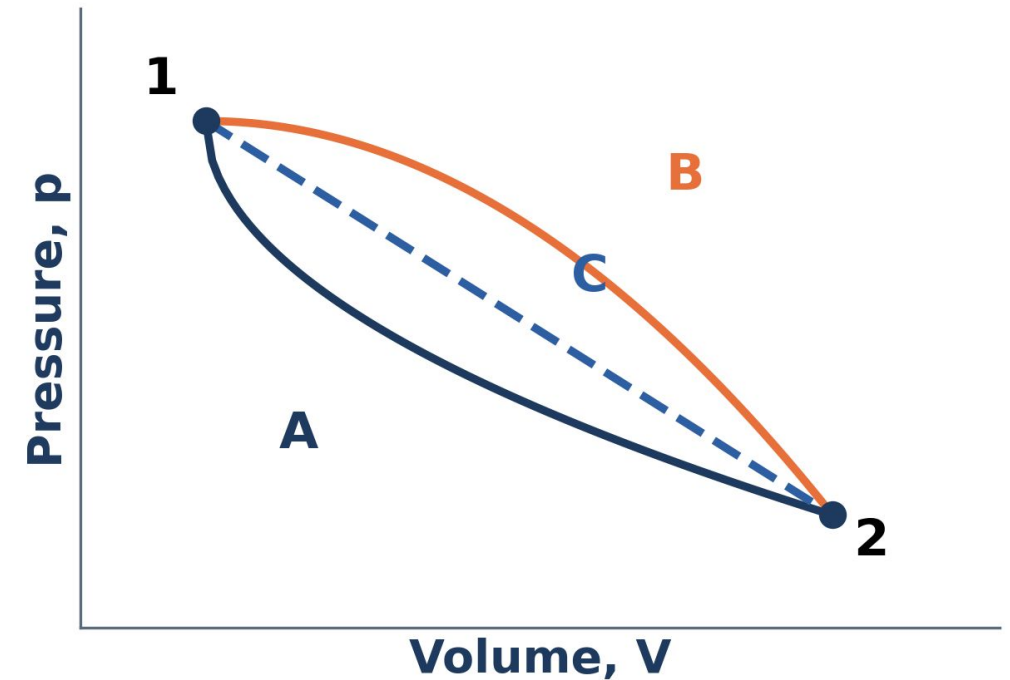
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Energy is a property

The first law for a process

Q – W depends only on the end states

- Go 1 → 2 by path A, return by path B: $(Q_A - W_A) + (Q_B - W_B) = 0$
- Return instead by path C: $(Q_A - W_A) + (Q_C - W_C) = 0$
- Subtract: $Q_B - W_B = Q_C - W_C$
- Q – W is the same for **every path** between the same states – it is the change of a **property**, the energy E



Two return paths B and C give the same Q – W

The first law for a process

$$Q - W = \Delta E = E_2 - E_1, \quad E = U + KE + PE$$

Stationary system: $Q - W = \Delta U \quad \delta Q - \delta W = dU$

- Heat added minus work done is stored as an increase in the energy of the system
- Energy is conserved – it can only be converted from one form to another

Corollaries of the first law

Internal energy is a property

$Q - W$ has the same value for every path between two states, so it is the change of a property of the system.

Energy of an isolated system is constant

With $Q = 0$ and $W = 0$, $\Delta E = 0$ – energy can be neither created nor destroyed, only converted.

A PMM1 is impossible

No device can deliver work continuously without an energy supply – it would create energy.

Perpetual motion machine of the first kind (PMM1)

- A device that would **deliver work continuously without any energy supply**
- It would create energy, violating the first law – impossible
- The converse is equally impossible: a machine that continuously absorbs work without any other effect

KEY POINT Every engine needs a fuel, a heat source or some other energy input.

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Internal energy

Internal energy U

- Energy stored by virtue of the **motion and configuration of the molecules**: translational, rotational and vibrational kinetic energy plus the potential energy of intermolecular forces
- Distinct from the macroscopic kinetic and potential energy of the system as a whole
- An **extensive property** and a **point function**: $\Delta U = Q - W$ depends only on the end states; $\oint dU = 0$
- For an ideal gas U depends on **temperature alone** (Joule's law); $u = U/m$ in kJ/kg

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Limitations of the first law

What the first law cannot tell us

- No information about the **direction** of a process – it would allow heat to flow from cold to hot
- No distinction between heat and work – yet heat cannot be fully converted to work in a cycle
- Nothing about whether a process is possible, or how far it proceeds
- Nothing about the **quality** of energy – how much of it can produce work

KEY POINT These limitations are removed by the second law (Unit II).

Summary

- Joule: work supplied $\propto$ heat rejected over a cycle; $4.186 \text{ kJ} = 1 \text{ kcal}$
- First law for a cycle: $\oint \delta Q = \oint \delta W$
- $Q - W$ is the same for every path $\Rightarrow$ energy E is a property
- First law for a process: $Q - W = \Delta E$; stationary system $Q - W = \Delta U$
- Energy of an isolated system is constant; PMM1 is impossible
- Internal energy: molecular energy; extensive; point function; $U = f(T)$ for an ideal gas
- First law gives no direction, no quality, no equilibrium conditions

Next: Enthalpy, specific heats and the first law applied to processes

- Enthalpy $H = U + pV$ and constant-pressure heating
- Specific heats c_v and c_p ; $c_p - c_v = R$
- First-law table for the standard non-flow processes; multi-process cycles
- Numerical problems: Worked Examples 1.1 – 1.5 in the lecture notes